



## High Performance Vision-Guided Rocket Tracker

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**TEAM #3 SPACEY | MCMASTER UNIVERSITY SFWRENG 4G06 | MCMASTER ROCKETRY TEAM**







## — THE CHALLENGE

# Rockets reach **Mach 3** in seconds.

Manual optical tracking is physically impossible for small-profile hobbyist rocketry. Commercial rigs cost upwards of \$10,000.

**100** km

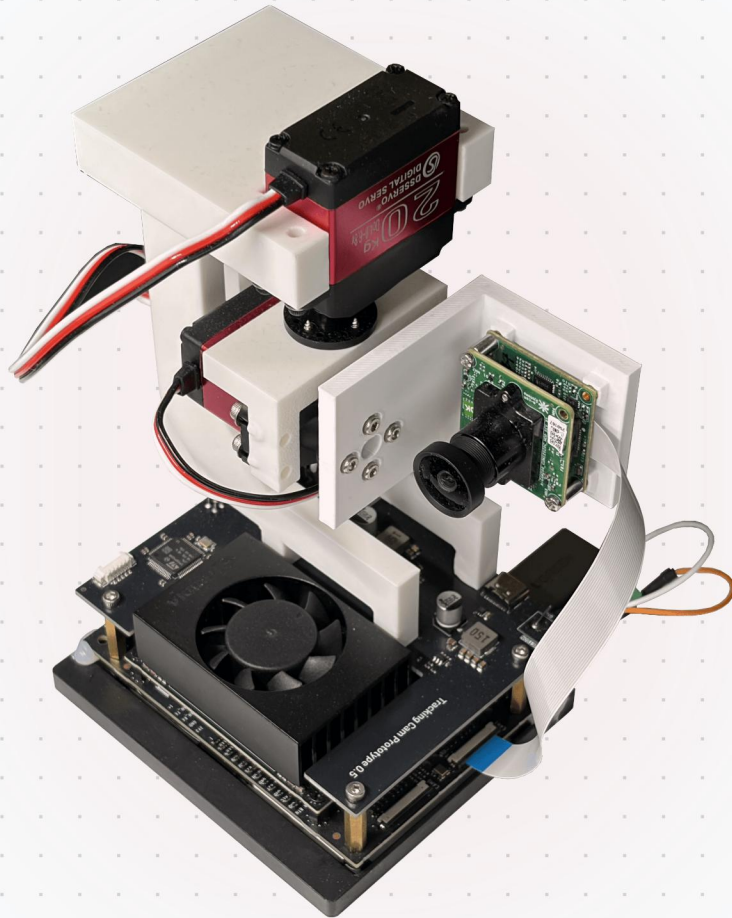
MAX ALTITUDE

**>10** g

ACCELERATION

## PRESENTATION ROADMAP

- 1. Problem & Solution
- 2. Architecture & Live Demo
- 3. Data, CV & ML Pipeline
- 4. Hardware & UI Control
- 5. Testing, Challenges & Future



# Autonomous. Accessible. Accurate.

A fully autonomous, closed-loop optical tracking system engineered to meet strict performance constraints.



## Autonomous Edge AI Tracking

Works in remote, offline, and hazardous conditions



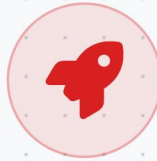
## Accurate and Precise

60Hz closed-loop control and digital stabilization



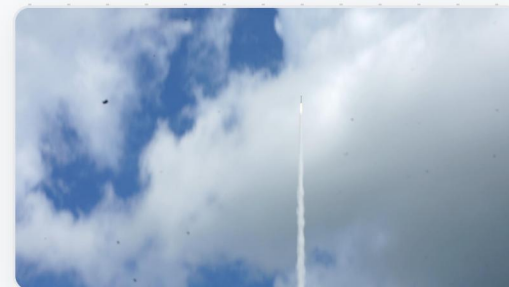
## Low Cost

Total BOM under \$500 CAD (vs \$10k+ commercial)

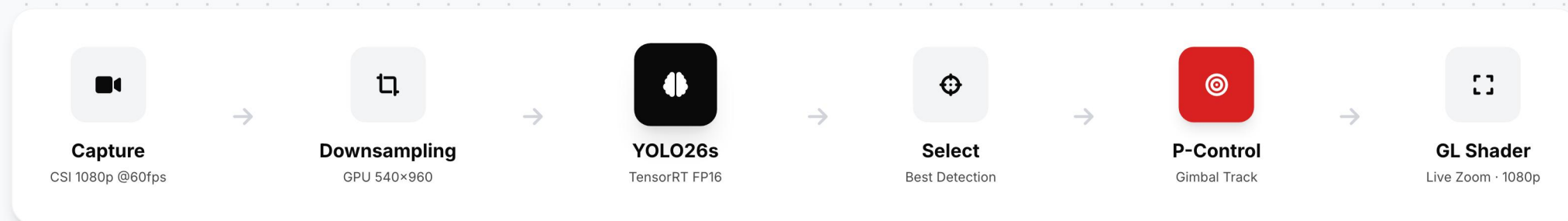


# LIVE DEMO

## Optimized for extreme distance and strict latency budgets.



Raw frame — can you spot the rocket?



### 🔍 Why YOLO26s?

One-to-one detection head eliminates NMS entirely. 9.47M parameters engineered for FP16—only 9% accuracy loss on Jetson vs. 30% for alternative architectures.

### 🔥 Full GPU Pipeline

DeepStream runs decode, downsampling, and TensorRT inference entirely on GPU. Confidence threshold filters noise; probe selects single best detection per frame.

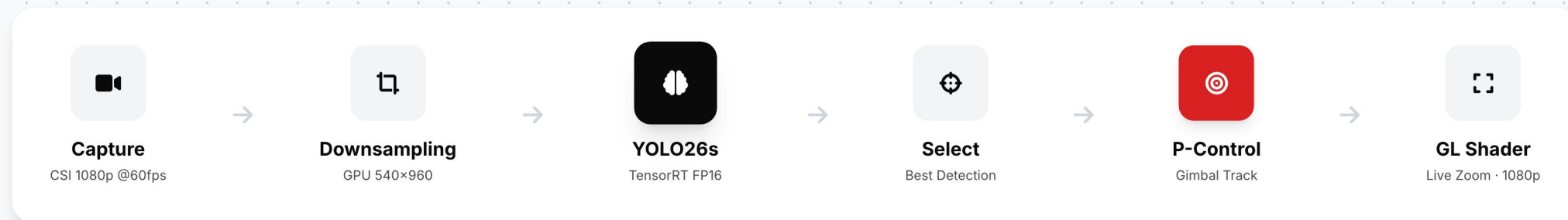
### 🎯 Instant Visual Lock

GL fragment shader crops and zooms on target within one display frame. P-control drives gimbal via UART. Zero perceived tracking latency.

## Optimized for extreme distance and strict latency budgets.



Target <0.1% of frame — detected at conf 0.52



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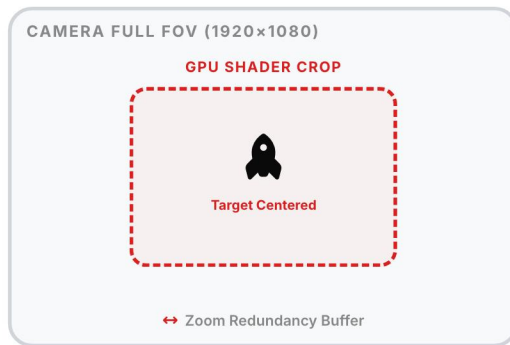
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# Optical redundancy enables perceptually instant tracking.



## FOV Redundancy Model

Wide optical view guarantees target stays in frame while gimbal mechanically catches up



## Detection Result

BBox (cx, cy, w, h) from YOLO @ 60fps

~1MS



## GPU Shader Path

- 1 Compute tx, ty, scale from bbox
- 2 GL shader crops & centers frame
- 3 Livestream output → operator sees instant lock

~100MS



## Gimbal Servo Path

- 1 P-control → delta pan/tilt
- 2 UART → STM32 → PWM servo
- 3 Mechanical motion re-centers optical FOV



## Zero Perceived Latency

Operator sees target locked within one display frame (~16ms). Gimbal catches up transparently.



# Custom labelling platform for rocket data.

All training data flows through our in-house **RoCam Labeler** — a full-stack annotation tool built from scratch to handle our unique dataset requirements.



## SAM-Assisted Annotation

One-click segmentation masks via Meta SAM



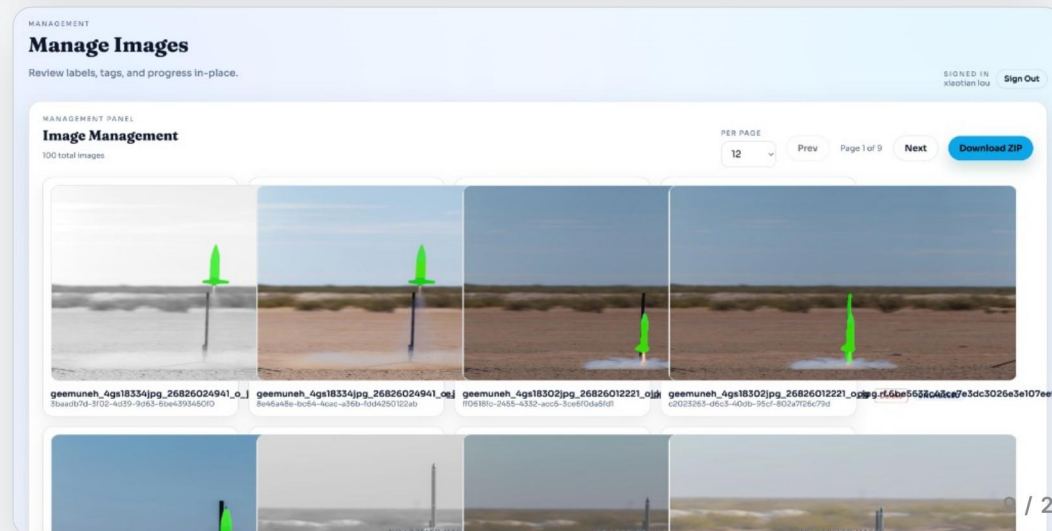
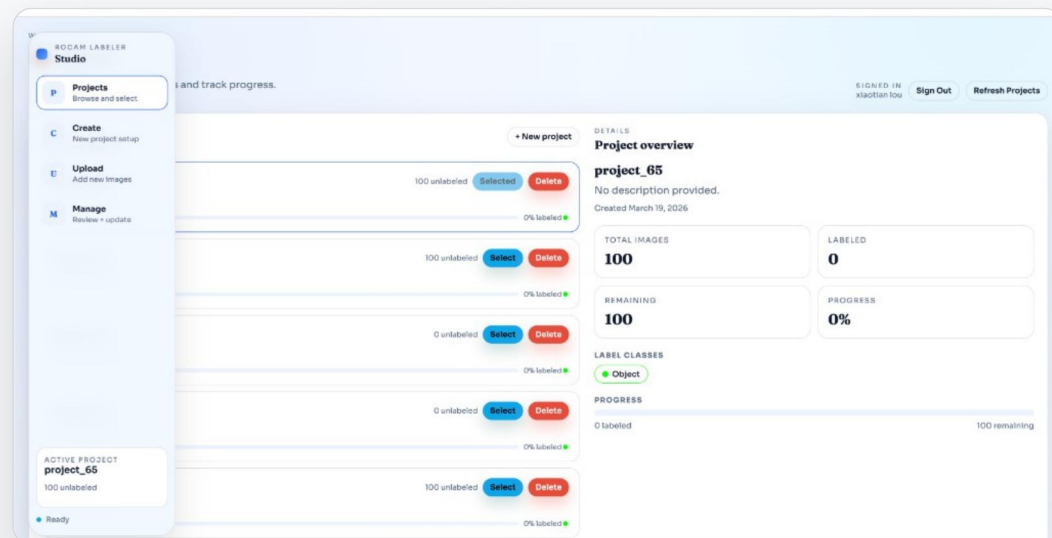
## Project Management

Multi-project workspace with progress tracking



## Batch Upload / Export

ZIP-based bulk import and YOLO-format export



# 96.2%

### PEAK TRAINING ACCURACY

mAP@50 across 10+ experiments on H100

## 9% Deployment Gap

FP16 quantization on Jetson Orin Nano. Our best-in-training architecture lost **30%** at deployment—the production recipe held.

**94.3%**  
PRECISION

**89.8%**  
RECALL

**60**  
FPS PIPELINE

**9.47M**  
PARAMETERS

**20.5**  
GFLOPS

**<1ms**  
SHADER LATENCY

## Aggressive Augmentation Training

YOLO26s trained end-to-end for **420 epochs** on H100 GPU. Data labelled via our custom **Labelling App** with SAM-assisted annotation.



### Full Mosaic = 1.0 (Key Finding)

Every image trained as 2×2 mosaic, quadrupling effective object density. Disabled last 80 epochs for full-resolution refinement. All reduced-mosaic variants failed.



### 15,832 Images + 4,000 COCO Hard Negatives

25% of training set is pure background from COCO, forcing a harder decision boundary to suppress false positives in real-world deployment.



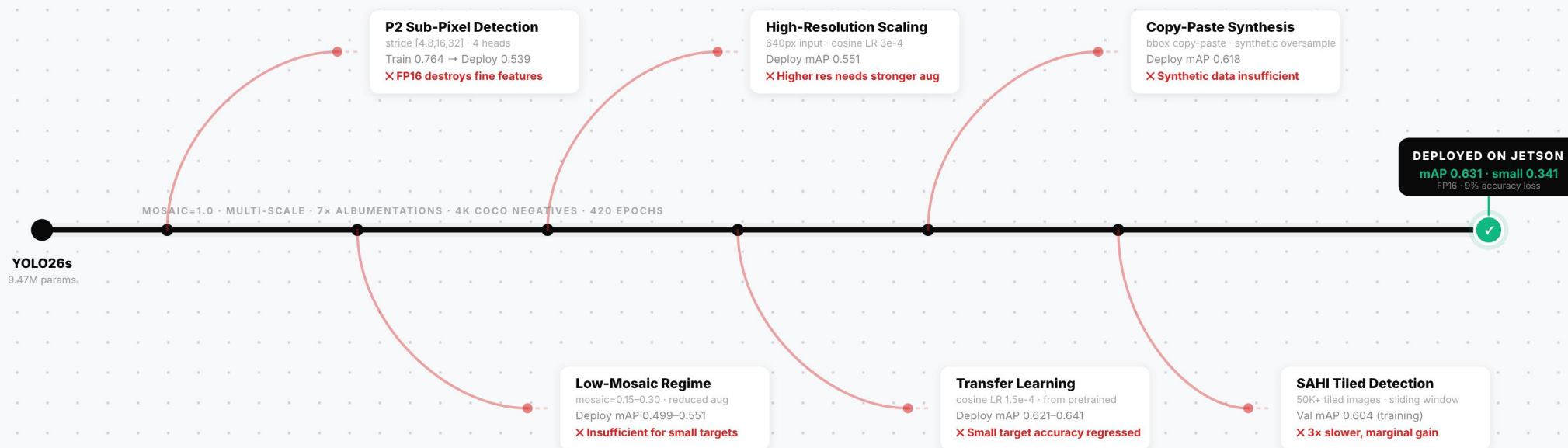
### Multi-Scale + 7 Albumentations

Variable resolution (544–960px), 180° rotation, motion blur, compression artefacts, and sensor noise transforms for real-world robustness.

## MODEL DEVELOPMENT JOURNEY

# Systematic Exploration. **One Survivor.**

Every promising architecture and training strategy tested against real edge deployment constraints



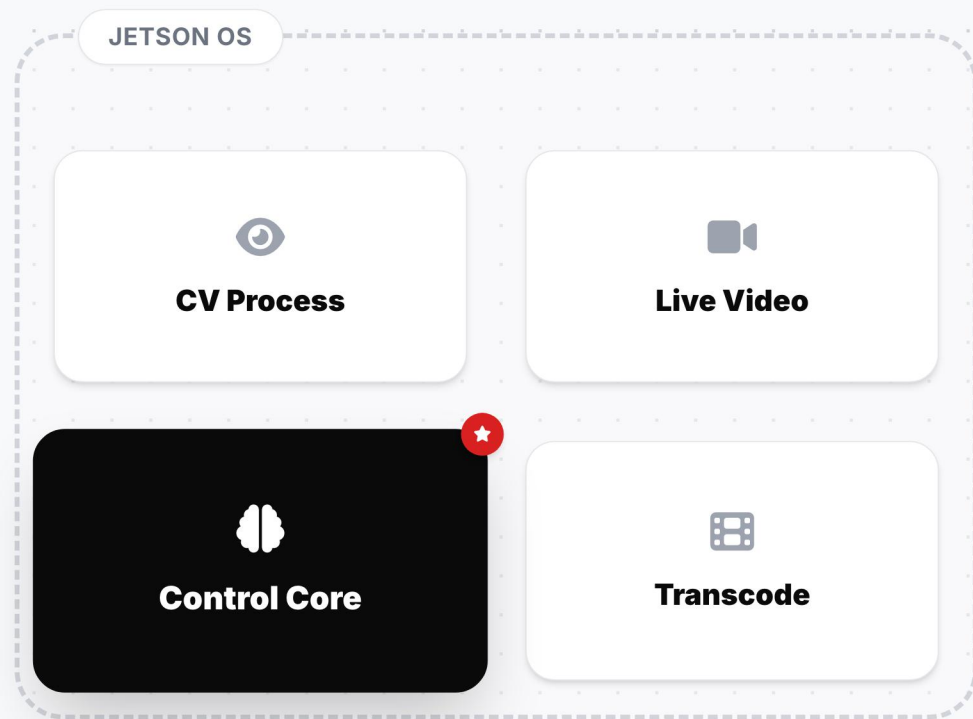
**10+** EXPERIMENTS  
CONDUCTED

**100+** GPU-HOURS  
ON H100

**9%** FP16 ACCURACY  
LOSS (BEST)

**1** SURVIVED  
DEPLOYMENT

## — FAULT-TOLERANT ARCHITECTURE



## Built for isolation and real-time survival.



### Independent Restart

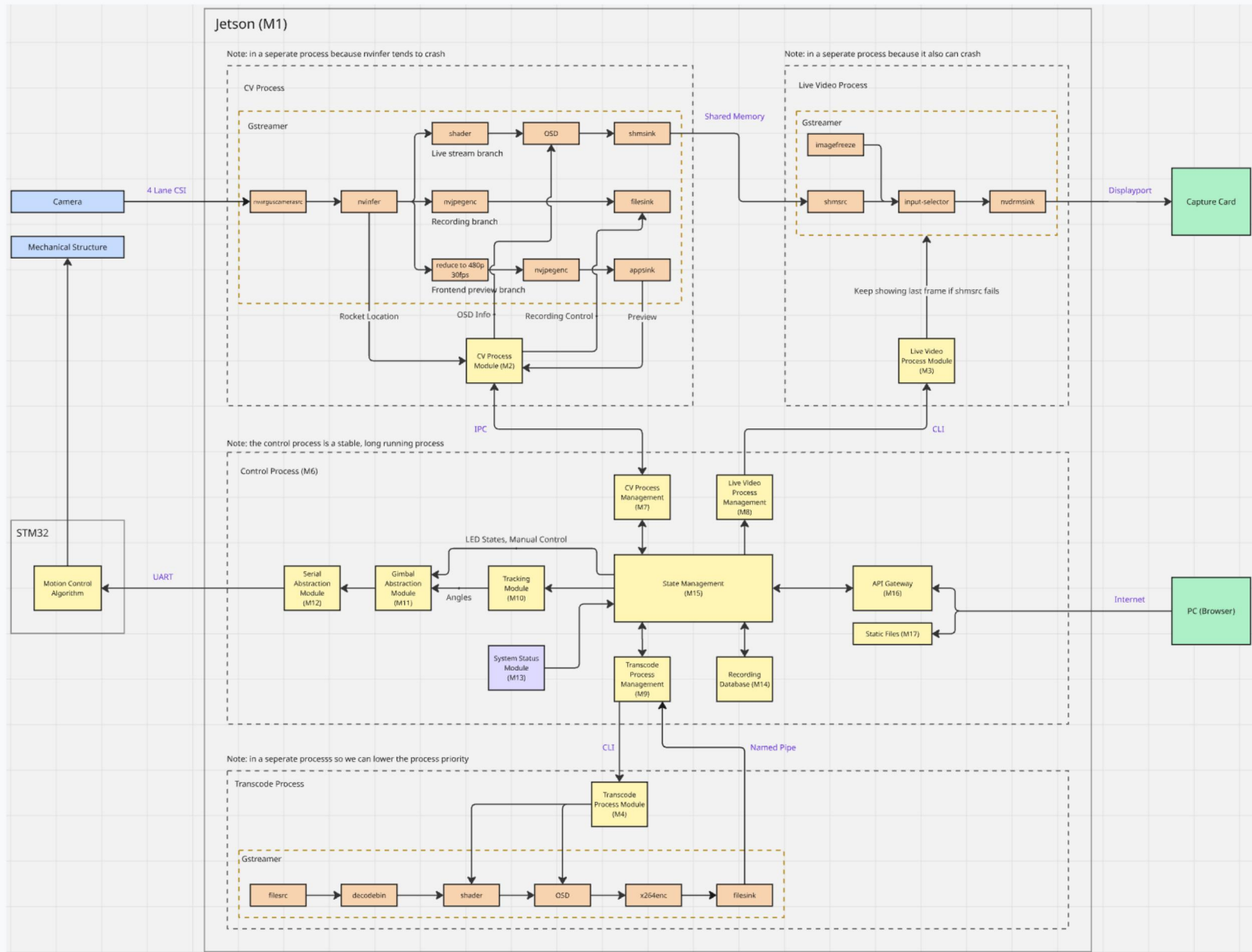
If the CV inference crashes, the control process maintains connection to the gimbal, preventing lock-up.



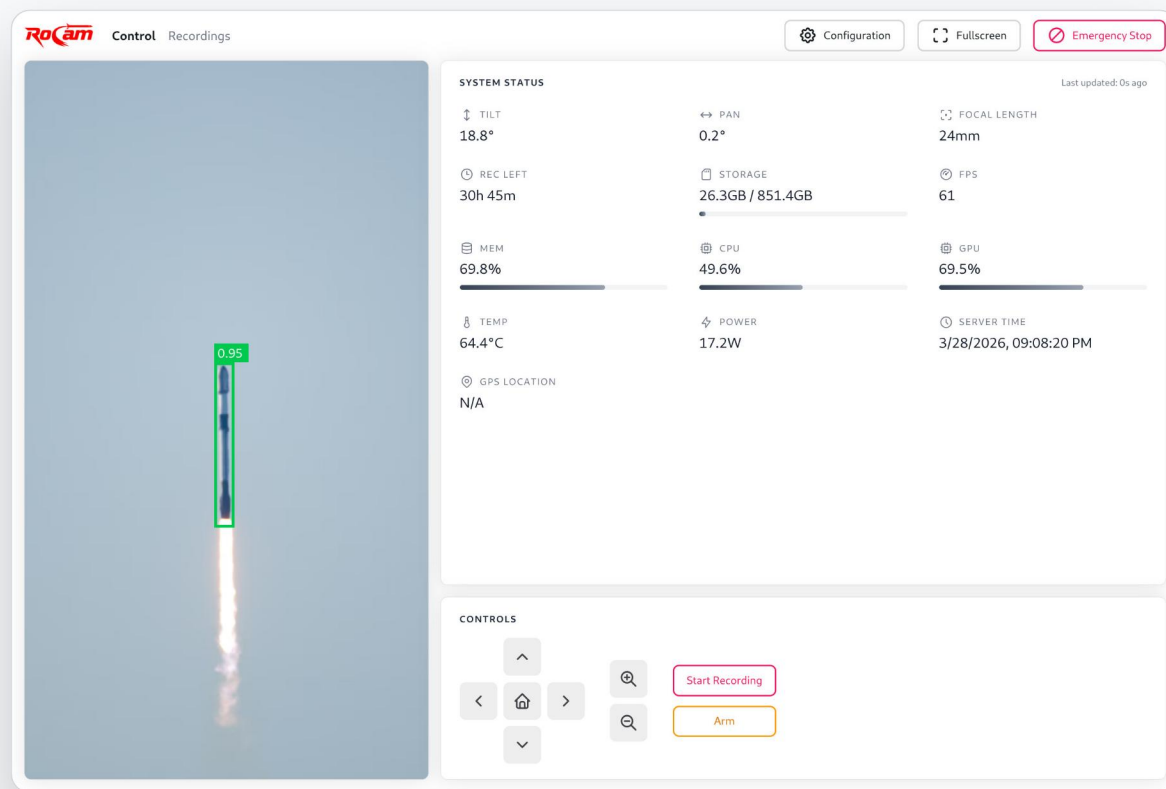
### Zero-copy IPC

Shared-memory architecture allows massive 1080p frames to move between processes instantly.





## MISSION CONTROL UI



## Command center in a browser.

REACT • TYPESCRIPT • VITE

📺 Low-latency live preview

⚙️ Direct manual gimbal override

📈 Real-time health telemetry

🌐 Full Bilingual (EN/FR) support

## — RECORDING WORKFLOW

### FROM LAUNCH TO REPLAY

# Record raw. Rebuild smart.

RoCam stores the raw camera feed and per-frame telemetry separately, then reconstructs stabilized playback and downloads on demand.

#### TRIGGER

##### Browser UI

Start / stop from  
Mission Control

#### STORAGE

##### AVI + JSONL

Video frames plus OSD  
telemetry

#### OUTPUT

##### Preview / Download

Rendered after the flight  
event

#### WHY THIS SPLIT MATTERS

-  Capture stays lightweight during launch instead of doing heavy replay rendering in the critical path.
-  The saved telemetry lets us re-apply OSD, timestamps, zoom transforms, and stabilization later for post-flight review.

#### 1. CAPTURE



### Raw video branch to **video.avi**

The CV pipeline swaps in a recording sink and writes the camera stream without waiting for replay rendering.



#### 2. LOG



### Per-frame OSD data to **log.txt**

Each frame stores pan, tilt, focal length, FPS, timestamps, and stabilization transforms as JSONL telemetry.



#### 3. REPLAY



### Separate transcode for stabilized review

Preview and download are reconstructed later with the same OSD and shader path, producing a polished replay stream for analysis.

# 481

## AUTOMATED TESTS PASSED



### Backend (pytest)

313 Unit Tests

**88% Coverage**



### Frontend (Vitest)

168 Component Tests

**86% Coverage**



### CI Pipeline

GitHub Actions on every PR

**100% Pass Rate**



## — PROJECT CHALLENGES

### **Real-time Latency**

Python's GIL blocking CV loops.



### **Solution**

Shared-memory IPC bypassing GIL entirely.

### **Micro-Object Detection**

Rockets vanishing into background noise.



### **Solution**

YOLO26s-P2 Head & Hard-negative mining.

### **Hardware Sync**

UART dropping packets at 115200 baud.



### **Solution**

Logic analyzer debug + custom CRC-8 checksums.

**Designed to scale from 200m hobby rockets to 3km+ high-power launches.**



### Performance

- 4K @ 60fps tracking
- Authentic ML Architecture
- Optical Auto-focus



### Hardware Scaling

- Full-size Gimbal Mounts
- Multi-camera Arrays
- Weatherproof Enclosures

NEXT STEP



### Software UX

- Mobile-native Apps
- Post-flight Analytics
- AR Telemetry Overlay

## Built with **operator feedback**, not just lab assumptions.

MCMMASTER ROCKETRY TEAM

**"A more direct and intuitive way to move the gimbal."**

Captured in stakeholder walkthrough. Turned into drag-to-move control on the live preview.



### Direct Control

Drag-to-move framing added beside the D-pad.



### Clear Status

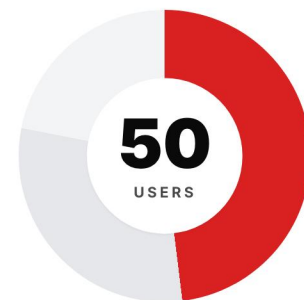
More visible armed, target-lost, storage, and network states.



### Stable Recording

Pause/resume behavior refined for download and recording.

MCMMASTER ROCKETRY TEAM · 50 MEMBERS



### Autonomous Tracking

24 members · 48%

### Manual Override

15 members · 30%

### Video Recording

11 members · 22%



# Thank You.

Questions?



<https://github.com/SpaceY-Labs/RoCam>

WITH THANKS TO



**SUPERVISOR**

**Dr. Shahin  
Sirouspour**

Faculty guidance and  
project support



**INSTRUCTOR**

**Dr. Spencer  
Smith**

Thanks to the course for the  
structure and direction.



**TEACHING ASSISTANT**

**Tanya  
Djavaheerpour**

Thanks for the feedback  
throughout the project.

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